

Mixed effects models

STAT340 - Lecture 12

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April 28 2026

Book chapters

- The R Book (Crawley): Chapter 19 (Mixed-Effects Models). Key sections:
 - Fixed and random effects
 - Pseudoreplication
 - Split-plot experiments
 - Nested designs
- An Introduction to Statistical Learning (ISLR):
 - *Note:* ISLR focuses mostly on standard i.i.d. predictive modeling. Mixed models go beyond the standard linear models discussed in ISLR Chapter 3, handling complex, non-independent data structures!

Recap: The standard i.i.d. assumption

In standard Linear Regression and Generalized Linear Models (GLMs), a core assumption is that observations are **independent and identically distributed (i.i.d.)**.

$$y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

- **Identically distributed:** Every observation y_i shares the same variance σ^2 and follows the same relationship with x_i .
- **Independent:** Knowing the residual (error) for observation y_1 tells us absolutely nothing about the residual for observation y_2 .

Why do standard i.i.d. models fail?

What if our data points are *grouped* or *structured*?

- Repeated measurements on the same patients over time.
- Plants growing in the same greenhouse.
- Students nested within classrooms.
- Counties epidemiology data may be spatially correlated.
- Time series data may be temporally correlated.

If we measure the same patient 5 times, those 5 measurements are **not independent**. If the patient is naturally prone to higher blood pressure than average, all 5 of their measurements will have a positive residual.

Why do standard i.i.d. models fail? (Cont.)

If we ignore this grouping structure and use a standard GLM:

1. The model assumes we have more independent information than we actually do (artificially inflated sample size).
2. Standard errors for our coefficients will be **underestimated**.
3. We will get **false positives** (p-values that are too small)!
4. Our estimates may be biased towards over-represented individuals' measurements.
5. i.i.d. does not hold when running the maximum likelihood estimation.

We need a model that explicitly accounts for this correlation structure.

Mixed effects models

- Mixed effects models are (generalized) linear models with both **fixed** and **random** effects.
- Often fixed treatments are studied with *replications* on randomly selected objects/locations.
 - Replications in time: Repeated measurements.
 - Spatial replication: Observations from close locations.
 - Spatio-temporal correlations.
- **Main issue:** Dealing with dependence in data.
- Mixed effect models account for dependency in data. Ignoring dependence can lead to flawed conclusions!

Fixed vs Random Effects (Crawley, R Book)

The distinction between fixed and random effects is crucial when building these models:

- **Fixed effects:** The levels of the factor are determined explicitly by the researcher. We are interested in the specific differences between these exact levels.
 - *Examples:* Drug vs Placebo, Male vs Female, High vs Low Temperature.
 - The conclusions drawn apply **only** to these specific levels.

Fixed vs Random Effects (Cont.)

- **Random effects:** The levels are a random sample from a much larger population of possible levels. We are not interested in the specific differences between the levels we sampled, but in the general variation they represent.
 - *Examples:* Individual patient ID, specific plots of land, specific litters of animals.

The Problem of Pseudoreplication

- Repeated measurements over time on the same «unit»
 - «Unit» = individual, group, location...etc.
 - Data from the same «unit» are similar to each other (dependent).
 - Also known as **longitudinal data**.
- Repeated measurements on the same «unit» are **pseudo-replicates**.
 - Not «real» replicates!
- We are not interested in the difference between each «unit», but rather see them as a **random sample** of a larger population of such «units».
 - Then the effects of «unit» are typically **random effects**.

The Problem of Pseudoreplication: Example

- **Scenario:** You want to test the effect of a new diet on milk yields in a herd of COWS.
- You have 10 cows in the herd.
- You measure the milk yield of each cow every day for 30 days.
- Is your sample size $n = 300$?
- **NO.** Your true sample size of independent units is $n = 10$ (the cows). The 30 daily measurements are **pseudoreplicates**! The daily measurements from “Cow 1” are highly dependent on Cow 1’s baseline genetics and health.

Split-Plot Experiments

Mixed models naturally handle complex experimental designs, such as **Split-Plot Designs**:

- **Main plots:** Large experimental units (e.g., entire fields) where a treatment is applied (e.g., Irrigation).
- **Sub-plots:** Smaller units within the main plots where another treatment is applied (e.g., Fertilizer type).
- **Why it matters:** The error variance for the main plot treatment is different (usually larger) than the error variance for the sub-plot treatment.
- Mixed models estimate variance components at each level of the hierarchy simultaneously!

Inference in Linear Mixed Models (High Level)

In standard regression, we use Maximum Likelihood Estimation (MLE) because we observe all the data. But in mixed models, the random effects (b_i) are **unobserved random variables**.

- We cannot just treat them as standard fixed parameters, or we'd overfit!
- Instead, the math **integrates them out**. We look at the “marginal likelihood” of the data given only the fixed effects and the variance parameters.
- For normal (Gaussian) data, this integration has a beautiful, exact mathematical solution.

MLE vs REML

When we estimate the variances using standard Maximum Likelihood (ML), the estimates are slightly biased (too small), because ML doesn't account for the degrees of freedom used up by estimating the fixed effects.

- Think of calculating sample variance: we divide by $n - 1$, not n , to fix the bias!
- Mixed models use **REML (Restricted Maximum Likelihood)** instead of ML to get unbiased estimates of the variance components (σ_b^2 and σ^2). This is the default in R!

Random Intercept Only Model

Let's start with the simplest mixed model: The **Random Intercept Model**.

This is equivalent to a One-Way ANOVA with random effects. We want to measure a continuous response across different random groups, without any specific fixed predictors (other than the global mean).

The rats dataset

Let's look at the `rats` dataset from the `BIAS.data` package.

We measure the Glycogen content in three different sections of rat livers. The rats themselves are randomly selected from a population.

```
library(BIAS.data)
data(rats)
head(rats, 7)
```

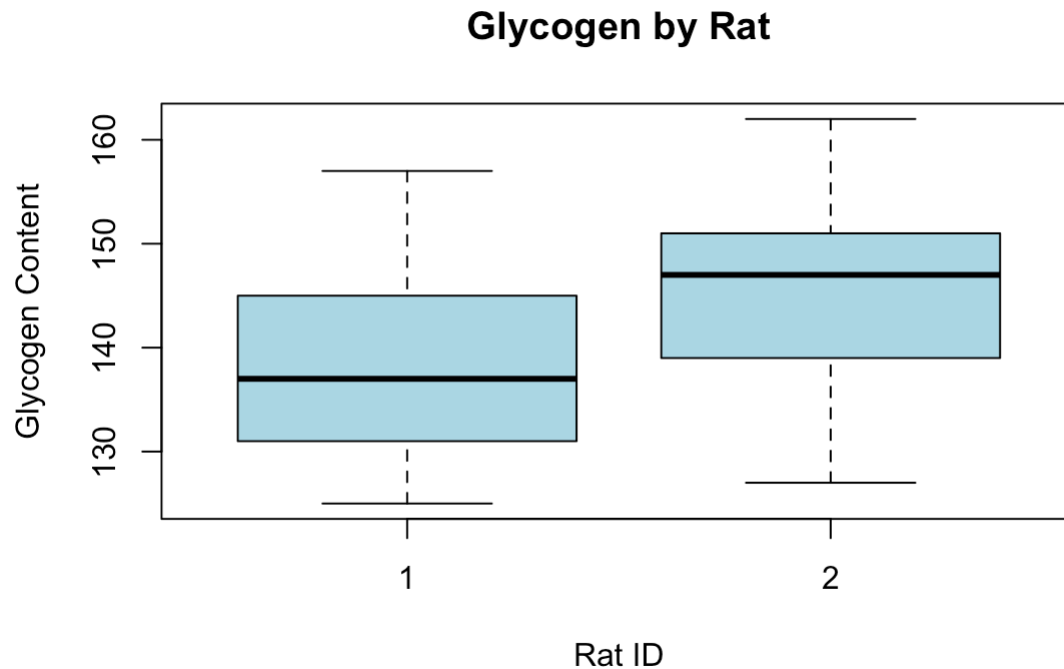
##	Glycogen	Treatment	Rat	Liver
## 1	131	1	1	1
## 2	130	1	1	1
## 3	131	1	1	2
## 4	125	1	1	2
## 5	136	1	1	3
## 6	142	1	1	3
## 7	150	1	2	1

Structure of the **rats** dataset

- Glycogen: The continuous response variable.
- Treatment: The section of the liver (1, 2, or 3).
- Rat: The specific rat the liver came from. This is our **random effect**!

Visualising the **rats** data

Let's look at the variance *between* rats versus the variance *within* rats.



Some rats naturally have higher baseline glycogen than others!

The Random Intercept Mathematical Model

To avoid complex matrix math, let's write the model for a single observation using scalar notation:

Let y_{ij} be the glycogen content for the j -th measurement of the i -th rat.

$$y_{ij} = \beta_0 + b_i + \epsilon_{ij}$$

Where:

- β_0 : The global intercept (average glycogen across all rats).
- b_i : The random effect for rat i . It represents how much rat i 's average deviates from the global average β_0 .
- ϵ_{ij} : The residual error for that specific measurement.

Model Assumptions

$$y_{ij} = \beta_0 + b_i + \epsilon_{ij}$$

What do we assume about the random terms?

1. **Random Effects:** $b_i \sim N(0, \sigma_b^2)$ The rat-specific deviations follow a Normal distribution with variance σ_b^2 .
2. **Residuals:** $\epsilon_{ij} \sim N(0, \sigma^2)$ The individual measurement errors follow a Normal distribution with variance σ^2 .
3. The random effects b_i and residuals ϵ_{ij} are completely independent.

Fitting the Random Intercept Model in R

We use the `lme()` function from the `nlme` package.

```
library(nlme)
# We fit a random intercept (~ 1) for each Rat
rat_model <- lme(fixed = Glycogen ~ 1,
                 random = ~ 1 | Rat,
                 data = rats)
```

Notice the syntax:

- `fixed = Glycogen ~ 1`: The only fixed effect is the global intercept.
- `random = ~ 1 | Rat`: We allow the intercept (**1**) to vary randomly across each Rat.

Full Summary: rats Model

```
## Linear mixed-effects model fit by REML
##   Data: rats
##       AIC       BIC    logLik
##  266.2975 270.9635 -130.1487
##
## Random effects:
## Formula: ~1 | Rat
##      (Intercept) Residual
## StdDev:    4.266496  9.26216
##
## Fixed effects:  Glycogen ~ 1
##              Value Std.Error DF   t-value p-value
## (Intercept) 142.2222  3.388876 34 41.96737      0
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -1.9334507 -0.6648481 -0.1658917  0.6068381  1.8854657
##
## Number of Observations: 36
## Number of Groups: 2
```

Explained Variance: The Intraclass Correlation Coefficient (ICC)

How much of the total variance in Glycogen is simply due to which rat we are looking at?

We calculate the **Intraclass Correlation Coefficient (ICC)**:

$$\text{ICC} = \frac{\text{Variance Between Groups}}{\text{Total Variance}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}$$

The ICC tells us the proportion of total variance that is explained by our random grouping factor. It also represents the correlation between two measurements taken from the *same* rat!

ICC Calculation for rats

From our output:

- $\sigma_b = 4.27$
- $\sigma = 9.26$

$$\sigma_b^2 = 4.27^2 \approx 18.23$$

$$\sigma^2 = 9.26^2 \approx 85.75$$

$$\text{ICC} = \frac{18.23}{18.23 + 85.75} \approx 0.175$$

Almost 17.5% of the variance in Glycogen is purely driven by biological differences between individual rats!

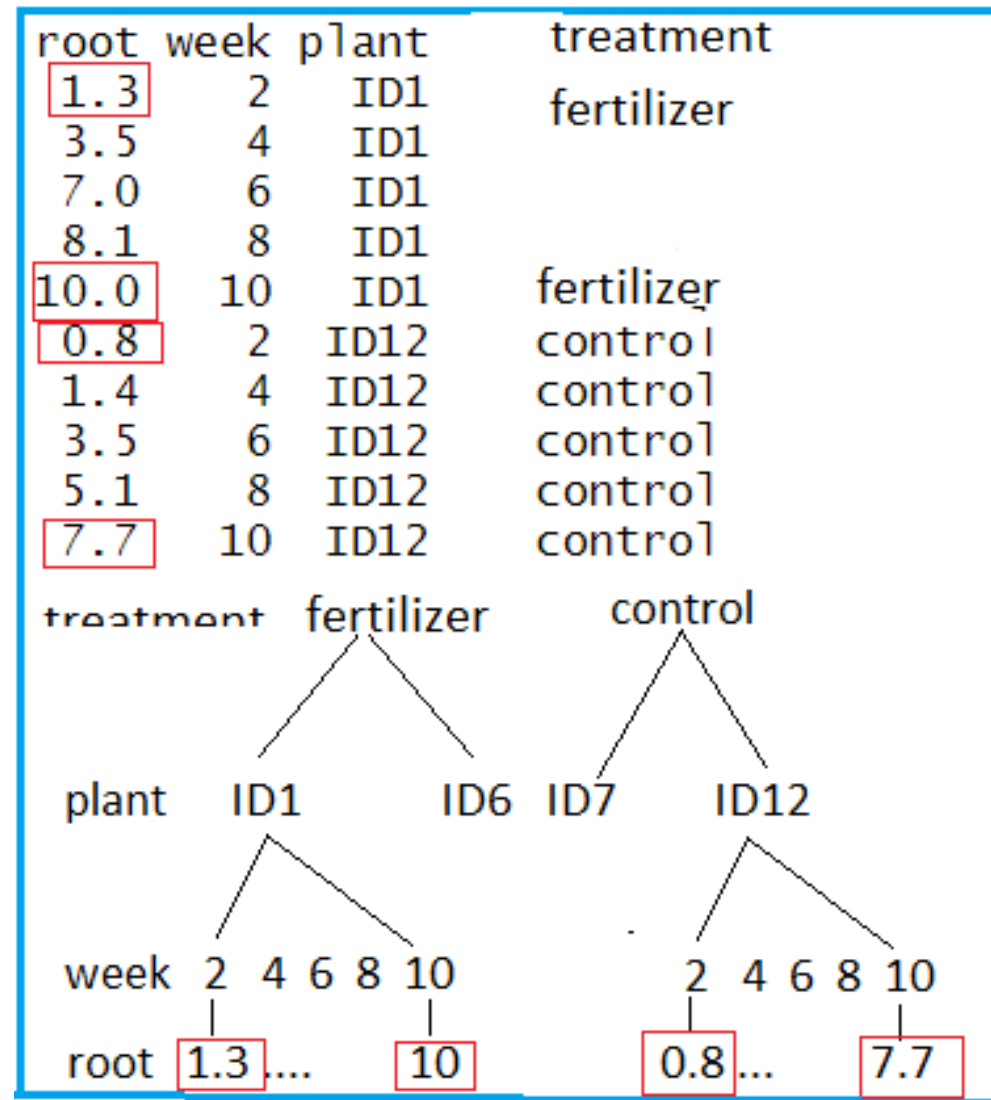
Moving to Random Slopes: The **fertilizer** dataset

What if a predictor variable (like time) affects different groups differently? Let's look at the **fertilizer** dataset.

```
print(fertilizer[c(3:8,48:53),])
```

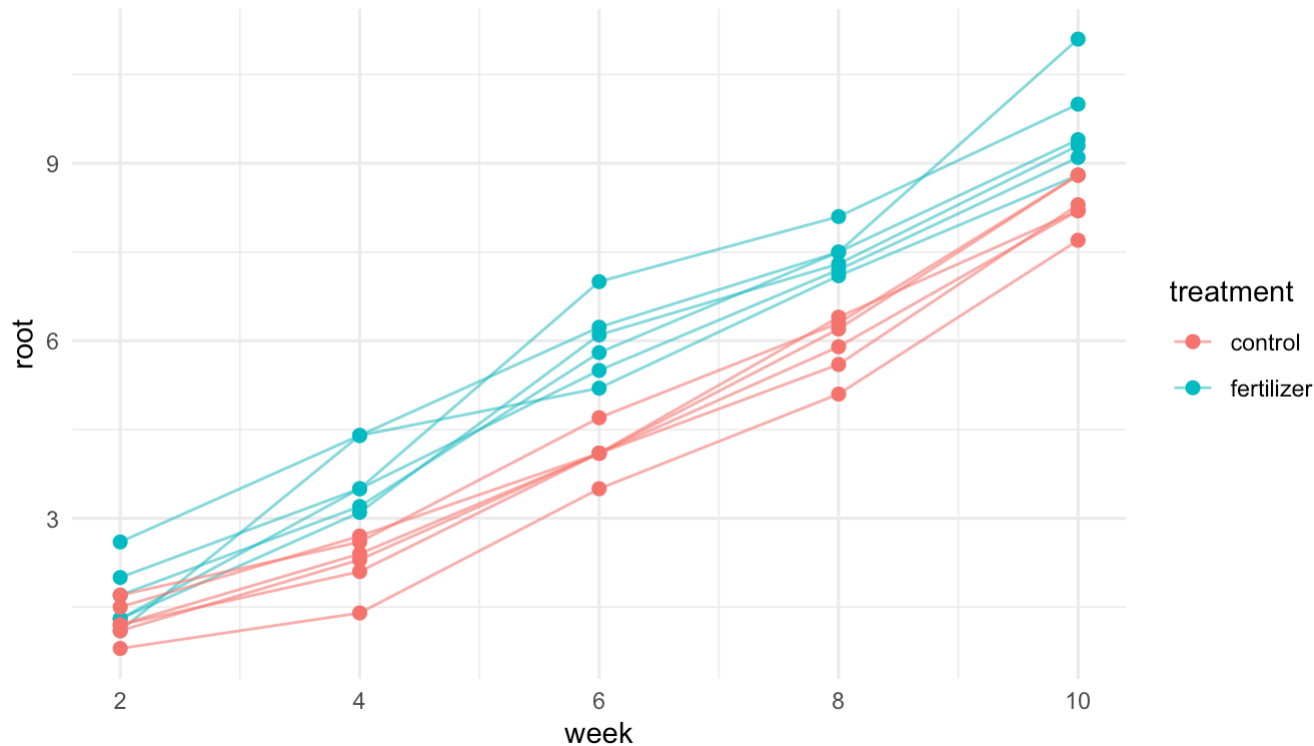
```
##      root week plant  treatment
## 3    7.0    6   ID1 fertilizer
## 4    8.1    8   ID1 fertilizer
## 5   10.0   10   ID1 fertilizer
## 6    2.0    2   ID2 fertilizer
## 7    3.5    4   ID2 fertilizer
## 8    5.5    6   ID2 fertilizer
## 48   4.1    6  ID10      control
## 49   6.2    8  ID10      control
## 50   8.8   10  ID10      control
## 51   1.2    2  ID11      control
## 52   2.4    4  ID11      control
## 53   4.1    6  ID11      control
```


Structure of **fertilizer**



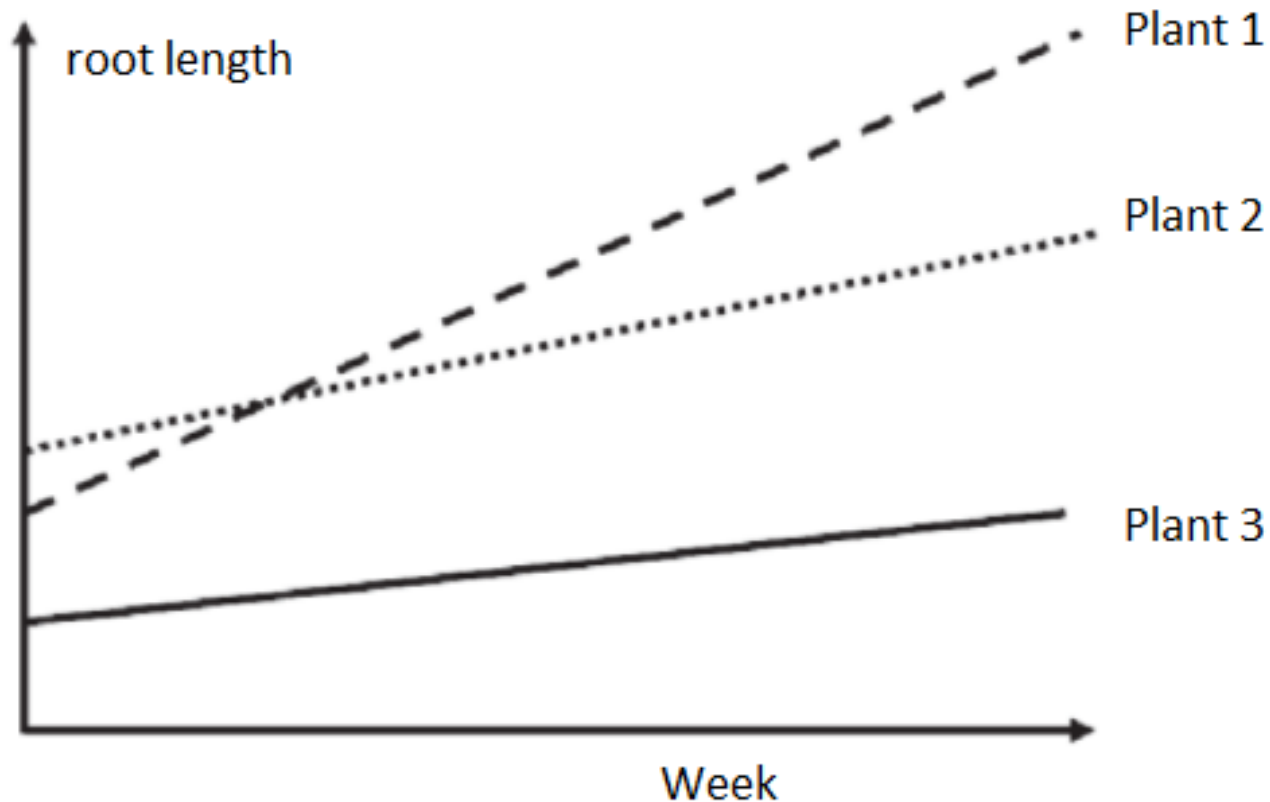
We track the **root** length of various **plants** over several **weeks**. The plants are either given **fertilizer** or are **controls**.

Visualising fertilizer trajectories



Look closely at the individual plants. They don't just start at different lengths (different intercepts); their lines may have different steepness (different slopes)!

Random intercept, random slope model



If we only used a random intercept, we would force all lines to be parallel. A **Random Slope** model allows each line to have its own angle!

The Random Intercept + Random Slope Mathematical Model

Let y_{ij} be the root length for the i -th plant at week x_{ij} .

$$y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})x_{ij} + \epsilon_{ij}$$

Where:

- β_0 : Global average intercept.
- β_1 : Global average slope (growth per week).
- b_{0i} : How much plant i 's intercept deviates from the average.
- b_{1i} : How much plant i 's slope deviates from the average.

Fitting the Random Slope Model in R

```
# Treatment is a fixed effect.  
# Both Intercept and Week slope vary randomly by Plant.  
res1 <- lme(fixed = root ~ treatment,  
            random = ~ week | plant,  
            data = fertilizer)
```

The syntax `random = ~ week | plant` tells R: “Allow the intercept AND the effect of `week` (slope) to vary randomly for each `plant`”.

Full Summary: fertilizer Model

```
## Linear mixed-effects model fit by REML
## Data: fertilizer
##      AIC      BIC    logLik
## 171.0236 183.3863 -79.51181
##
## Random effects:
## Formula: ~week | plant
## Structure: General positive-definite, Log-Cholesky parametrization
##           StdDev   Corr
## (Intercept) 2.8639832 (Intr)
## week        0.9369412 -0.999
## Residual    0.4966308
##
## Fixed effects: root ~ treatment
##           Value Std.Error DF   t-value p-value
## (Intercept)  1.760327 0.1438367 48 12.238371 0e+00
## treatmentfertilizer 1.039383 0.2034158 10  5.109644 5e-04
## Correlation:
##           (Intr)
## treatmentfertilizer -0.707
##
## Standardized Within-Group Residuals:
##           Min      Q1      Med      Q3      Max
## -1.9928118 -0.6586834 -0.1004301  0.6949714  2.0225381
##
## Number of Observations: 60
## Number of Groups: 12
```

summary(lme): Random Effects

Let's examine the random effects standard deviations:

- StdDev for (Intercept): $\sigma_{b0} \approx 2.86$.
- StdDev for week: $\sigma_{b1} \approx 0.94$.
- Residual standard error: $\sigma \approx 0.50$.

This tells us there is substantial variation between plants in both their starting size AND their growth speed.

ICC Calculation for **fertilizer**

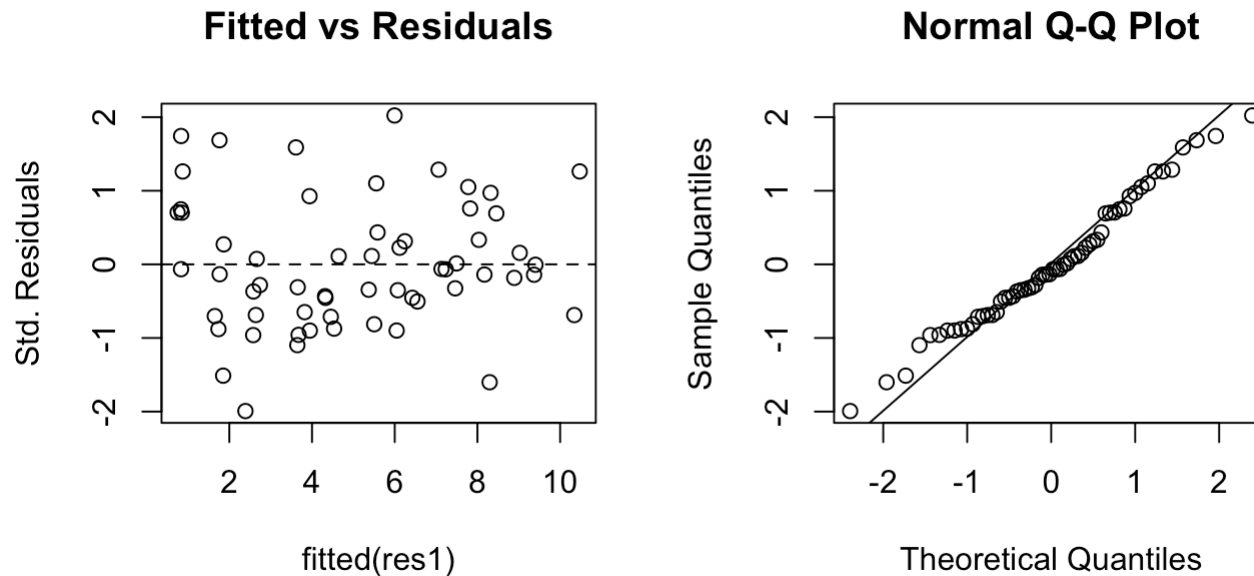
Because we have a **random slope** for **week**, the variance between plants *changes over time*. Therefore, the ICC also changes over time!

At week 0 (the starting point), the variance between plants is just the variance of the intercept σ_{b0}^2 :

$$\text{ICC}_{\text{week}=0} = \frac{\sigma_{b0}^2}{\sigma_{b0}^2 + \sigma^2} = \frac{2.86^2}{2.86^2 + 0.50^2} \approx 0.97$$

An ICC of 97% means that at the beginning of the experiment, almost all the variation in root length is driven by the specific plant's genetics/seed, rather than random noise!

Residual Diagnostics for **fertilizer**



What we want to see: Points randomly scattered around the zero line in the left plot, and falling nicely on the diagonal line in the right plot.

Looks ok!

summary(lme): Fixed Effects

Let's examine the fixed effects (the global averages):

```
round(summary(res1)$tTable, 4)
```

##	Value	Std.Error	DF	t-value	p-value
## (Intercept)	1.7603	0.1438	48	12.2384	0e+00
## treatmentfertilizer	1.0394	0.2034	10	5.1096	5e-04

summary(lme): Fixed Effects Intuition

From the table:

- **(Intercept)**: The baseline root length for the control group.
- **treatmentfertilizer = 1.04**: Plants with fertilizer are, on average, 1.04 units longer than control plants. This difference is significant ($p = 0.0005$).
- Notice that **week** is not in this fixed table because we only included it as a random slope. R estimates an average slope of 0 implicitly, but allows each plant to deviate from it.

Predictions: Population vs Individual

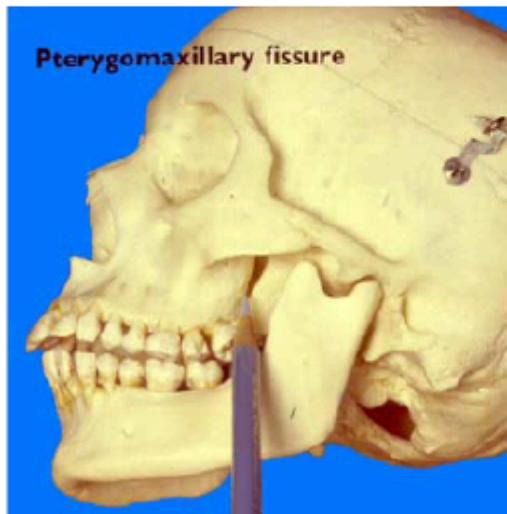
We can predict using our mixed model in two ways: - **Level 0 (Population)**: Uses only Fixed Effects (ignores random deviations). What is the average expected response? - **Level 1 (Individual)**: Adds the specific Random Effects for each unit.

```
# Predict for the first 5 observations
pred_pop <- predict(res1, level = 0)[c(1,10,32,40)]
pred_ind <- predict(res1, level = 1)[c(1,10,32,40)]
data.frame(Week = fertilizer$week[c(1,10,32,40)],
            Actual=fertilizer$root[c(1,10,32,40)],
            Treat = fertilizer$treatment[c(1,10,32,40)],
            Pop=round(pred_pop, 2), Ind=round(pred_ind, 2))
```

##	Week	Actual	Treat	Pop	Ind
## ID1	2	1.3	fertilizer	2.80	1.65
## ID2	10	9.1	fertilizer	2.80	9.02
## ID7	4	2.6	control	1.76	2.74
## ID8	10	8.2	control	1.76	8.03

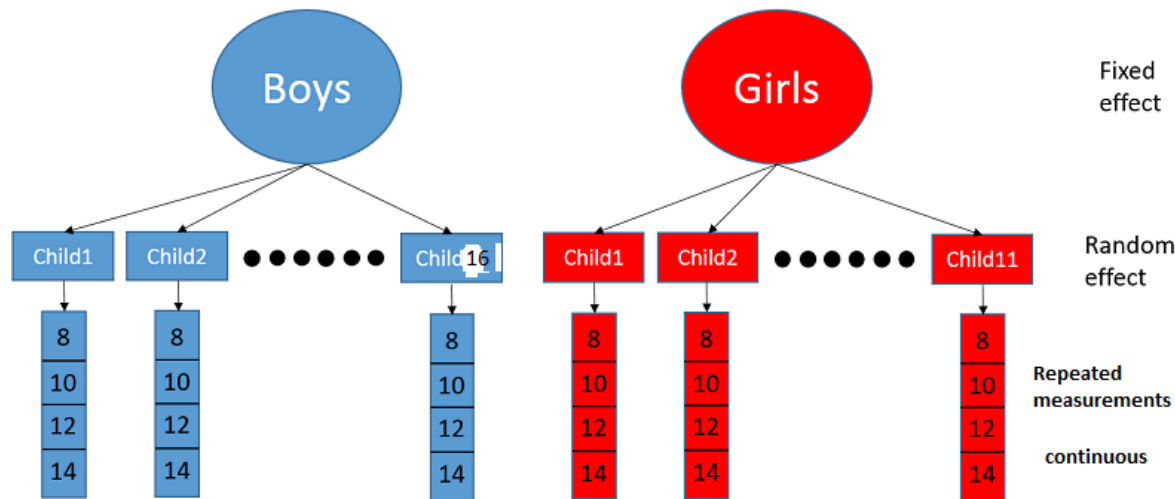
Orthodont data.

An example of renown - and distress, and misery. The distance from the pterygomaxillary fissure to the center of the pituitary gland is easily obtained, from xrays:



Hierarchical (Nested) Design: Orthodont Data

Sometimes random effects are nested within each other.



In the `Orthodont` dataset, we measure distance in children over time.

- `Child` is nested within `gender`.
- We have a fixed effect (`Sex`) and a random effect (`Subject` randomly chosen).

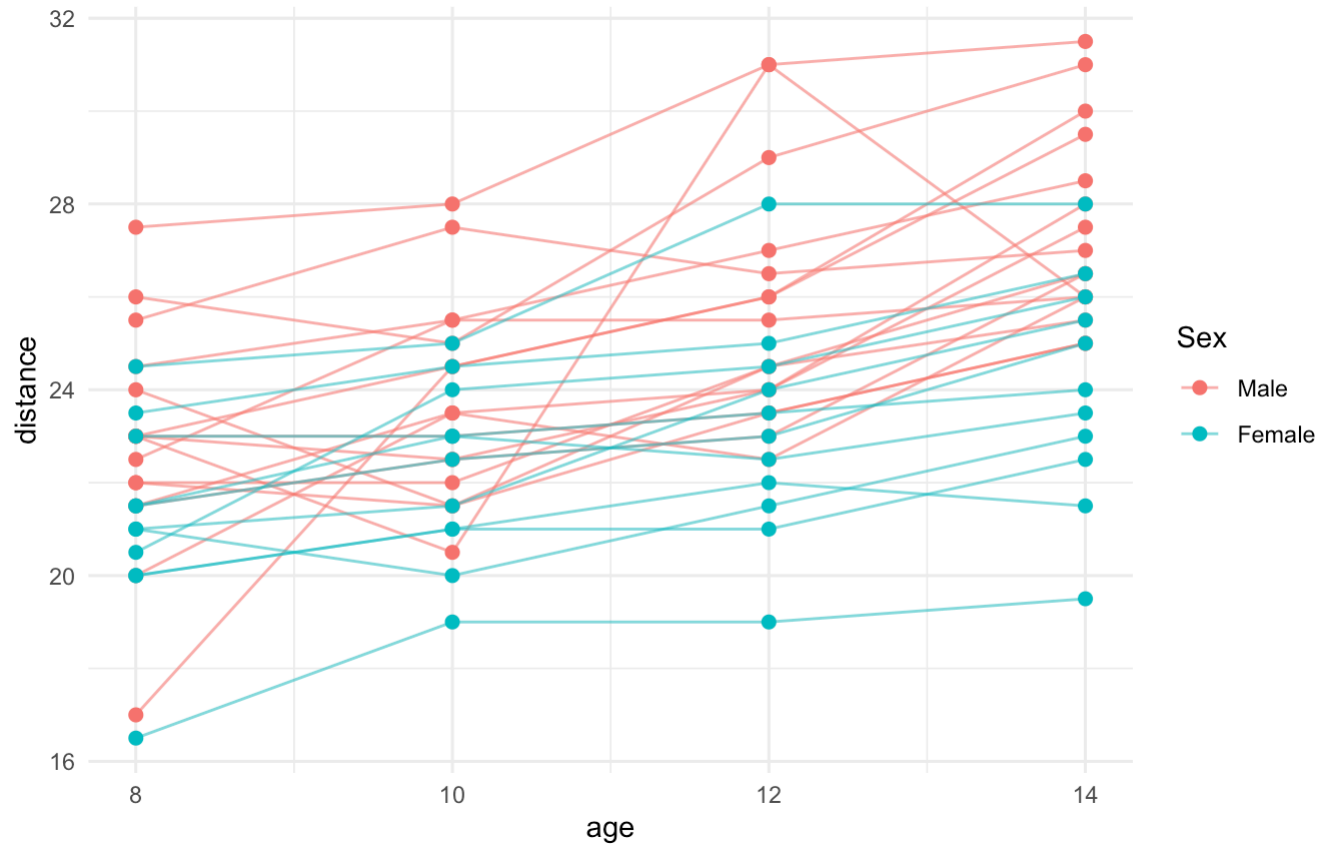
Structure of Orthodont

```
data(Orthodont)
head(Orthodont, 3)
```

```
## Grouped Data: distance ~ age | Subject
##   distance age Subject  Sex
## 1      26   8      M01 Male
## 2      25  10      M01 Male
## 3      29  12      M01 Male
```

We measure **distance** between some key points in the skull at different **ages**. **Subject** is the nested random effect.

Visualising Orthodont Data



Distances differ between girls and boys over time, and variance appears to increase slightly.

Orthodont Mathematical Model

Let y_{ijk} be the distance for the i -th subject, of the k -th sex, at the j -th age. Let's fit a random intercept model:

$$y_{ijk} = \beta_0 + \beta_1 \text{Age}_{ijk} + \beta_2 \text{Sex}_k + b_{i(k)} + \epsilon_{ijk}$$

- β_1 and β_2 are fixed effects.
- $b_{i(k)}$ is the random intercept for subject i nested within sex k .

Fitting **Orthodont** in R

```
res3 <- lme(fixed = distance ~ age + Sex,  
            random = ~ 1 | Subject,  
            data = Orthodont)
```

R automatically understands the nesting because each **Subject** has a unique ID (e.g., **M01**, **F01**) that maps exclusively to one **Sex**.

Full Summary: Orthodont Model

```
## Linear mixed-effects model fit by REML
## Data: Orthodont
##      AIC      BIC    logLik
## 447.5125 460.7823 -218.7563
##
## Random effects:
## Formula: ~1 | Subject
##      (Intercept) Residual
## StdDev:      1.807425 1.431592
##
## Fixed effects: distance ~ age + Sex
##              Value Std.Error DF   t-value p-value
## (Intercept) 17.706713 0.8339225 80 21.233044 0.0000
## age          0.660185 0.0616059 80 10.716263 0.0000
## SexFemale   -2.321023 0.7614168 25 -3.048294 0.0054
## Correlation:
##      (Intr) age
## age      -0.813
## SexFemale -0.372 0.000
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -3.74889609 -0.55034466 -0.02516628 0.45341781 3.65746539
##
## Number of Observations: 108
## Number of Groups: 27
```

Orthodont Fixed Effects Details

```
round(summary(res3)$tTable, 4)
```

##	Value	Std.Error	DF	t-value	p-value
## (Intercept)	17.7067	0.8339	80	21.2330	0.0000
## age	0.6602	0.0616	80	10.7163	0.0000
## SexFemale	-2.3210	0.7614	25	-3.0483	0.0054

Orthodont Fixed Effects Intuition

- **age**: For every year older, distance increases by 0.66 units.
- **SexFemale**: Females have a baseline distance that is 2.32 units **lower** than males.
- Both are highly significant.
- Because this is a random intercept model, the 0.66 growth slope is assumed to be exactly the same for every child. Only their baseline starting intercepts differ!

ICC Calculation for **Orthodont**

We fit a random intercept model for the **Orthodont** data:

$$\textit{distance} \sim \textit{age} + \textit{Sex} + (1|\textit{Subject})$$

From the `summary(lme)` output, we extract the variance components: - Subject variance (Between-child): $\sigma_b^2 = 1.807^2 \approx 3.267$ - Residual variance (Within-child): $\sigma^2 = 1.432^2 \approx 2.049$

$$\text{ICC} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2} = \frac{3.267}{3.267 + 2.049} \approx 0.615$$

Conclusion: **61.5%** of the variance in pituitary distance (after accounting for age and sex) is due to permanent biological differences between individual children.

Adding random slope to **Orthodont**

```
res4 <- lme(fixed = distance ~ age + Sex,  
            random = ~ age | Subject,  
            data = Orthodont)
```

Full Summary: Orthodont Model

```
## Linear mixed-effects model fit by REML
## Data: Orthodont
##      AIC      BIC    logLik
## 449.2339 467.8116 -217.6169
##
## Random effects:
## Formula: ~age | Subject
## Structure: General positive-definite, Log-Cholesky parametrization
##              StdDev   Corr
## (Intercept) 2.7970227 (Intr)
## age         0.2264274 -0.766
## Residual    1.3100398
##
## Fixed effects: distance ~ age + Sex
##              Value Std.Error DF   t-value p-value
## (Intercept) 17.635200 0.8862449 80 19.898788  0.000
## age         0.660185 0.0712532 80  9.265338  0.000
## SexFemale   -2.145492 0.7574539 25 -2.832504  0.009
## Correlation:
##              (Intr) age
## age         -0.838
## SexFemale   -0.348  0.000
##
## Standardized Within-Group Residuals:
##              Min      Q1      Med      Q3      Max
## -3.08141704 -0.45675583  0.01552695  0.44704158  3.89437694
##
## Number of Observations: 108
## Number of Groups: 27
```


Generalized Linear Mixed Models (GLMM) Intro

What if our response variable is **not continuous/normal**? For example, what if it's count data (Poisson) or binary data (Binomial), but with random effects?

We use **Generalized Linear Mixed Models (GLMM)**! The most popular package in R for this is `lme4`, specifically the `glmer()` function.

Inference in GLMMs (High Level)

With normal data (LMMs), integrating out the random effects has an exact, clean mathematical solution.

With non-normal data (like Binomial or Poisson), **the integral has no exact mathematical solution!** We cannot just use REML.

We must rely on approximations to calculate the likelihood:

- **Laplace Approximation:** Very fast, standard default in `lme4`. Uses a quadratic Taylor expansion to approximate the integral.
- **Gauss-Hermite Quadrature:** More computationally intensive, but highly accurate numeric integration.

GLMM - Formulation

Like standard GLMs, responses y_{ij} are assumed to follow a distribution parameterized by the mean μ_{ij} and dispersion ϕ :

$$y_{ij} \sim f(\mu_{ij}, \phi)$$

We model the expected value $\mu_{ij} = E(y_{ij})$ using a link function $g(\mu)$ while adding our random effects b_i :

$$g(\mu_{ij}) = \beta_0 + \beta_1 x_{ij} + b_i$$

- Fixed Effects: β_0 (Intercept) and β_1 (Slope)
- Random Effects: $b_i \sim N(0, \sigma_b^2)$

GLMM: Bovine Pleuropneumonia

Let's use the `cbpp` dataset in `lme4`. We want to model the incidence of a disease (Binomial) over different periods, accounting for the random effect of the specific herd.

```
library(lme4)
data(cbpp)
head(cbpp, 6)
```

```
##   herd incidence size period
## 1    1         2   14      1
## 2    1         3   12      2
## 3    1         4    9      3
## 4    1         0    5      4
## 5    2         3   22      1
## 6    2         1   18      2
```

cbpp dataset structure

- `incidence`: The number of sick cows (Successes).
- `size`: Total cows in the herd (Trials).
- `period`: The time period (Fixed effect).
- `herd`: The specific herd (Random effect).

Because the incidence rate might be naturally higher in some herds than others due to environment, **herd** must be a random effect!

Fitting the GLMM (`glmer`)

In `lme4`, we combine fixed and random effects in a single formula. Random effects are specified in parentheses: `(1 | herd)`.

```
# Response is cbind(successes, failures)
glmm_mod <- glmer(cbind(incidence, size - incidence) ~
                  period + (1 | herd),
                  family = binomial(link="logit"),
                  data = cbpp)
```

- `(1 | herd)`: A random intercept for each herd.
- `family = binomial`: Tells R to use Binomial regression and a logit link!

Full Summary: glmer Model

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: cbind(incidence, size - incidence) ~ period + (1 | herd)
## Data: cbpp
##
##          AIC          BIC      logLik -2*log(L)  df.resid
##        194.1        204.2       -92.0     184.1       51
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.3816 -0.7889 -0.2026  0.5142  2.8791
##
## Random effects:
## Groups Name          Variance Std.Dev.
## herd   (Intercept) 0.4123   0.6421
## Number of obs: 56, groups: herd, 15
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.3983     0.2312  -6.048 1.47e-09 ***
## period2      -0.9919     0.3032  -3.272 0.001068 **
## period3      -1.1282     0.3228  -3.495 0.000474 ***
## period4      -1.5797     0.4220  -3.743 0.000182 ***
## ———
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) perid2 perid3
## period2 -0.363
## period3 -0.340  0.280
## period4 -0.260  0.213  0.198
```

glmer Output Detail

```
round(summary(glm_mod)$coefficients, 4)
```

##	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	-1.3983	0.2312	-6.0478	0.0000
## period2	-0.9919	0.3032	-3.2721	0.0011
## period3	-1.1282	0.3228	-3.4948	0.0005
## period4	-1.5797	0.4220	-3.7430	0.0002

- The fixed effects show how the log-odds of incidence change across periods compared to period 1. The disease rate is decreasing over time (negative coefficients).

Spatiotemporal Correlation Intro

Sometimes, the “random effect” structure isn’t just discrete groups (like `plant` or `herd`). Observations might be correlated in **continuous time** or **across 2D space**.

If measurements are taken sequentially in time, residuals are often auto-correlated (an observation is similar to the one immediately before it).

In `nlme`, we can add explicit **correlation structures** without necessarily using random effects, or in combination with them.

Fitting Temporal Autocorrelation (AR1)

If we measure the same object over time at equal intervals, we can use an AR(1) process for the residuals. Let's fit this to the **Orthodont** data.

Using lme to add an CAR(1) correlation structure across age

```
model_car1 <- lme(fixed = distance ~ age + Sex,  
                  random = ~ 1 | Subject,  
                  correlation = corCAR1(form = ~ 1 + age | Subject),  
                  data = Orthodont)
```

Full Summary: corCAR1 Model

```
## Linear mixed-effects model fit by REML
## Data: Orthodont
##      AIC      BIC    logLik
## 449.3968 465.3206 -218.6984
##
## Random effects:
## Formula: ~1 | Subject
##      (Intercept) Residual
## StdDev:      1.788899 1.454494
##
## Correlation Structure: Continuous AR(1)
## Formula: ~1 + age | Subject
## Parameter estimate(s):
##      Phi
## 0.2418536
## Fixed effects: distance ~ age + Sex
##      Value Std.Error DF   t-value p-value
## (Intercept) 17.721416 0.8500193 80 20.848250 0.0000
## age          0.659405 0.0634074 80 10.399499 0.0000
## SexFemale    -2.327485 0.7611852 25 -3.057711 0.0053
## Correlation:
##      (Intr) age
## age      -0.821
## SexFemale -0.365 0.000
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -3.683026668 -0.540915318 -0.008097445 0.461167542 3.612579065
##
## Number of Observations: 108
## Number of Groups: 27
```

corAR1 Output Details

Let's look at the correlation structure parameters:

```
summary(model_car1)$modelStruct$corStruct
```

```
## Correlation structure of class corCAR1 representing  
##      Phi  
## 0.2418536
```

- Phi (Φ) is roughly 0.24.
- This represents the strength of the temporal correlation between measurements for the same child!

Spatial Correlation Intro

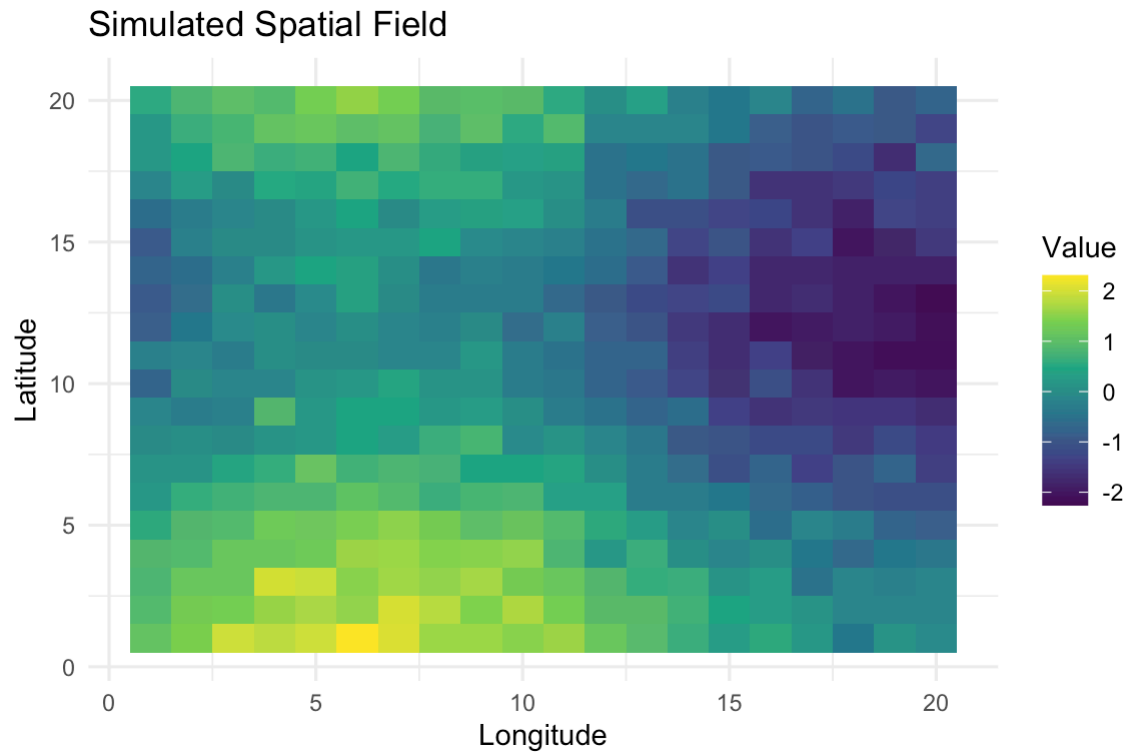
If data are collected across geographical coordinates (x, y) , points closer together are usually more similar.

Intuition: A measurement of soil pH at one spot is likely very similar to a measurement taken just 1 meter away!

We can model this spatial decay using exponential, spherical, or Gaussian correlation structures.

```
# Example syntax for a spatial model in nlme
spatial_model <- gls(yield ~ fertilizer + temperature,
                     correlation = corExp(form = ~ x_coord + y_coord,
                                          nugget = TRUE),
                     data = spatial_data)
```

Visualizing Spatial Correlation



Values that are close together in 2D space are similar in color. A mixed model uses spatial correlation to properly model this!

Summary

- Standard i.i.d. models fail when data is grouped or structured (Pseudoreplication).
- Mixed models combine Fixed effects (specific treatments) and Random effects (sampled populations).
- **Inference:** Uses REML (Linear) or Laplace approximations (GLMMs) because random effects are unobserved.
- `nlme::lme()` allows complex random intercepts/slopes and correlation structures (e.g. `corCAR1`).

STAT340: The Complete Journey

Over the past semester, we have built a powerful, flexible statistical toolkit. The course can be divided into distinct statistical domains:

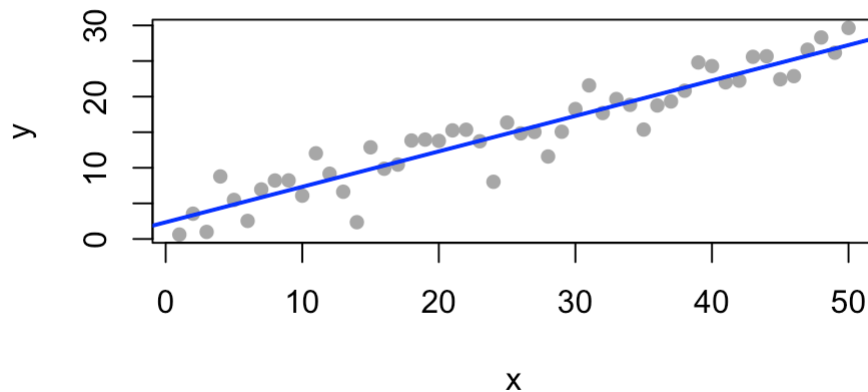
- **Lectures 1:** Introduction
- **Lectures 2-5:** Linear Models & ANOVA (Continuous data)
- **Lectures 6-7:** Clustering & classification
- **Lectures 8-9:** Dimension Reduction (PCA, PCR, PLSR)
- **Lectures 10-11:** Generalized Linear Models (GLM)
- **Lecture 12:** Mixed Effects Models (LMM & GLMM)
- **Guest Lecture 1:** *Explainable Bayesian Deep Learning through Latent Binary Bayesian Neural Networks*
- **Guest Lecture:** *A Smooth Information Criterion for Selection and Fusion of Categorical Predictors*

Linear Regression (Lectures 1-3)

The core framework: we model a continuous, normally distributed response y as a linear combination of predictors X .

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \epsilon_i \quad \text{where} \quad \epsilon_i \sim N(0, \sigma^2)$$

Simple Linear Regression

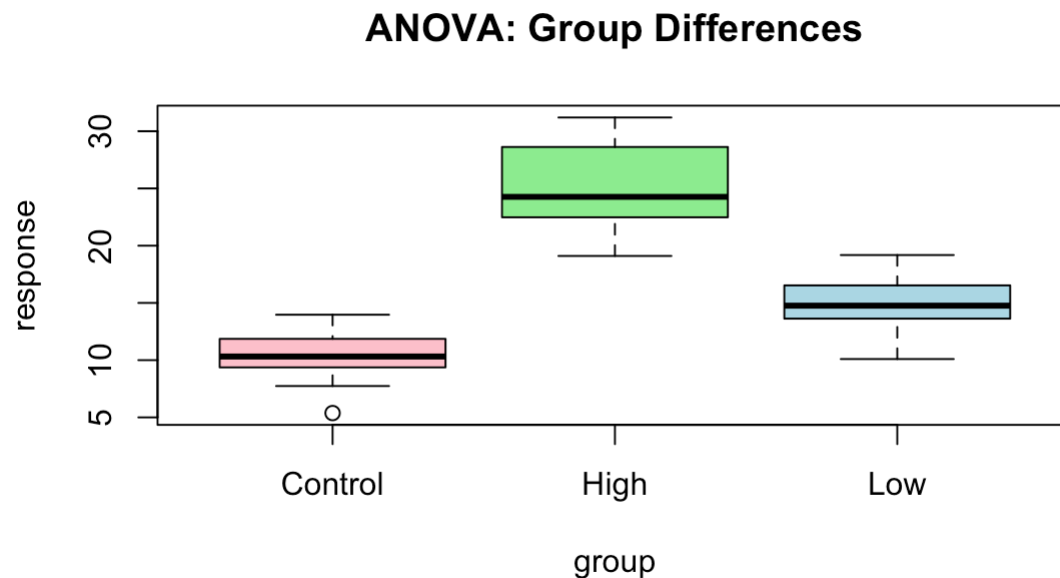


Key Assumption: Observations are independent, Gaussian and identically distributed (i.i.d.).

ANOVA & Categorical Predictors (Lectures 4-5)

Linear models seamlessly handle categorical predictors (like Treatments or Diets) by converting them into dummy variables.

$$y_{ij} = \mu + \alpha_j + \epsilon_{ij}$$

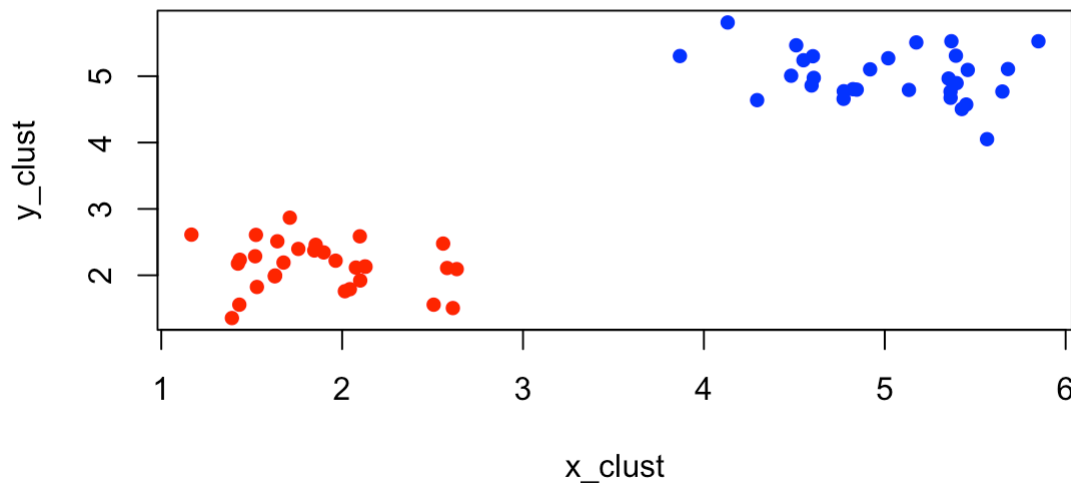


We used F-tests and Tukey HSD to compare means across groups!

Unsupervised Learning: Clustering (Lecture 6)

- **K-Means:** Partitioning data into K distinct groups by minimizing within-cluster distance.
- **Hierarchical Clustering:** Building a tree of clusters based on distance linkages (Single, Complete, Average).

K-Means Intuition



Classification: KNN (Lecture 7)

K-Nearest Neighbors (KNN) is a simple, non-parametric classification algorithm.

To predict the class of a new observation, we: 1. Find the K closest training points (using Euclidean or Manhattan distance). 2. Take a majority vote!

- **Small K** (e.g., $K = 1$): Highly flexible, captures complex boundaries, but prone to over-fitting.
- **Large K** : Smoother boundaries, less sensitive to noise.

Also

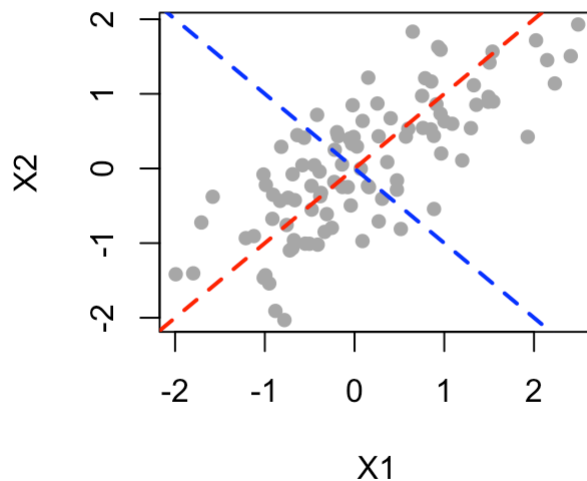
- **LDA** for linear discrimination
- **QDA** for quadratic discrimination

Dimension Reduction: PCA (Lecture 8)

When we have many correlated predictors, we face **multicollinearity**. **Principal Component Analysis (PCA)** solves this by rotating the data into orthogonal (uncorrelated) components.

- **PC1**: The direction of maximum variance.
- **PC2**: The direction of maximum variance orthogonal to PC1.

PCA Intuition: Finding Maximum Variance



PCR & PLSR (Lecture 9)

We can use PCA components as predictors in a regression model!

- **Principal Component Regression (PCR):** Extract PCs from X , then run linear regression on y .
 - *Problem:* PCA ignores y ! The components with highest variance might not be the most predictive of y .
- **Partial Least Squares Regression (PLSR):** Extracts components from X that *maximize covariance* with y . It is supervised dimension reduction!

Generalized Linear Models (Lectures 10-11)

What if y_i is **not** continuous and normal? We use a **Link Function** $g(\mu)$ to map the expected value μ to the linear predictors!

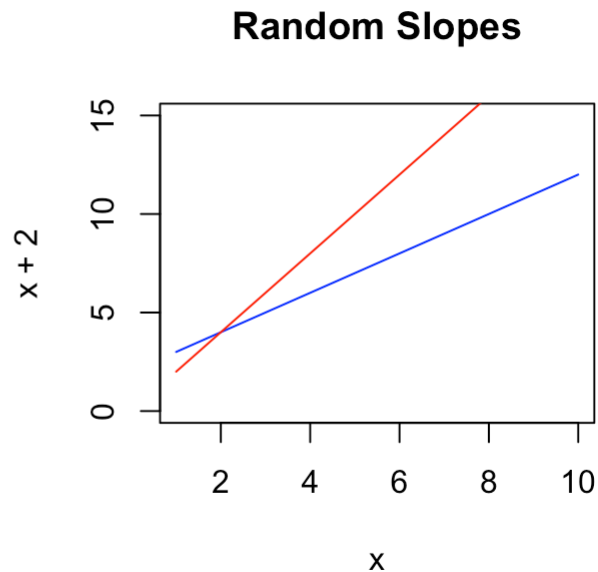
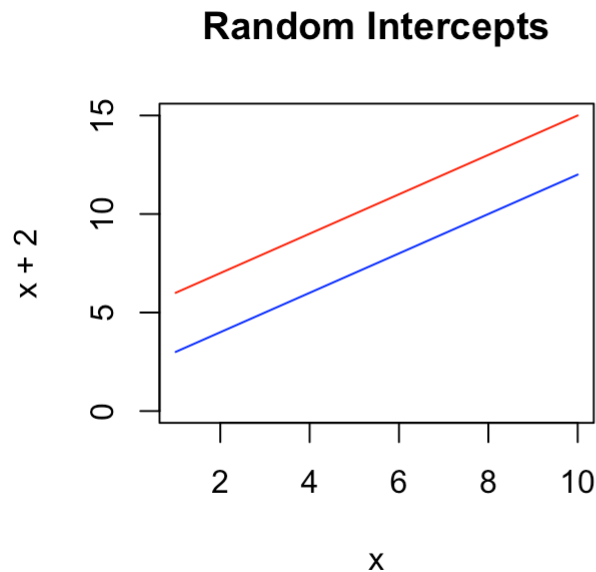
$$g(\mu_i) = \beta_0 + \beta_1 x_{1i} + \dots$$

Common Distributions & Links:

- **Normal (Continuous):** Identity link
- **Binomial (Binary 0/1):** Logit link
- **Poisson (Counts):** Log link
- **Gamma (Positive Skewed):** Log link

Mixed Effects Models (Lecture 12)

- **Random Intercept:** Lines have different starting points, but parallel angles.
- **Random Slope:** Lines have different angles based on group traits.



The Golden Rule of Modeling

There is no single “correct” model.

1. **Check the Response:** Is it continuous, binary, or count? (Choose LM vs GLM link).
2. **Check the Predictors:** Are there too many correlated predictors? (Use PCA/PLSR/model selection).
3. **Check the Structure:** Are observations truly independent, or nested/grouped? (Choose fixed vs mixed effects).

Thank you for a great semester in STAT340!

Good luck at the exam!